

# Preliminary Mitacs Intern Research Project Plan

**Title (Initial):** Hardware Accelerated YOLO on PYNQ for Smart Traffic Systems

**Date:** June-August 2026

**Supervision:** Prof. Seokbum Ko (Supervisor), Ebrahim Fard (PhD student)

## 1. Objective

This project falls within a specialized research area focused on hardware-accelerated deployment of deep neural networks (DNNs) for real-time computer vision on embedded platforms. It explores performance-optimized inference of object detection models, such as YOLO, on the PYNQ-Z2 FPGA-based system by combining high-level Python development with low-level hardware acceleration. The research integrates embedded AI and hardware-software co-design, with a focus on practical applications such as intelligent traffic monitoring and control. The project aims to demonstrate how resource-constrained edge devices can be used effectively for real-time, on-device inference in real-world scenarios.

## 2. Phases

A preliminary project phasing, including stages, is as follows. Notably, some stages may have been completed at the time the project begins. Hence, considering work balance, new stages may be added accordingly.

### Phase 1) Familiarization & Baseline

- Week 1: Review of the literature to check the state-of-the-art works
- Week 2: Toolchain setup and workflow refresh
- Week 3: Baseline inference on an embedded platform
- Week 4: Profiling and bottleneck identification

### Phase 2) Co-design & optimization

- Week 5 & 6: Model-level optimization (e.g., quantization, pruning, simplification)
- Week 7: Hardware-software partitioning
- Week 8: FPGA-based acceleration using overlays

### Phase 3) Evaluation

- Week 9: Latency and throughput measurements
- Week 10: Resource utilization analysis
- Week 11: Comparison against software-only baseline
- Week 12: Preliminary power or efficiency discussion

## 3. Deliverables by the end of the internship

- Month 1: System-level architecture and block diagrams
- Month 2: Functional FPGA-accelerated inference pipeline
- Month 3: Experimental results covering: performance, accuracy impact, resource usage, and a structured paper draft including figures and tables for the IEEE Transactions

## **4. Student preparation (Before start date)**

- Review recent papers on NN accelerators and embedded inference
- Refresh FPGA knowledge (e.g., Vivado/Vitis, HLS workflows)
- Practice quantized NN inference on FPGA-based platforms

## **5. Outcome**

The technical contributions of this project are expected to form the basis of at least one manuscript for the IEEE Transactions, to be prepared in Fall 2026, with the PhD student and two intern students as the first, second, and third authors, respectively, followed by Prof. Seokbum Ko as the corresponding author.